Summer Quarter
2026

Data Center Impact Explorer

Launch Presentation

Team JAMA
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Project Goals

1

Investigate relationship between data centers and local communities on multiple axes: environment, health, and politics

2

Aggregate numerous disparate sources of data to gain a more holistic understanding

3

Clean and preprocess data to facilitate analysis, understanding, and data visualization

4

Create an interactive dashboard linking data center data to political, environmental, and health data

Team Members

Anna Grace

Bachelor's in Math and Computer Science, worked in the telecom field doing financial analysis.

I will be examining data centers' potential impact on the power grid.

Morgan Murphy

Bachelor's in Psychological Science and certificate in Management Information Systems, worked in Human Factors research studying healthcare technology.

I will be exploring how data centers may impact health.

Jillian Kunze

Bachelor's in Physics and Astronomy, worked in science communication at a professional society for applied mathematicians.

I will be investigating the interplay between data center siting and local political leanings.

Andrea MacGregor

Bachelor's in Engineering and work as a Business Intelligence Analyst for a government consulting firm.

I will be exploring the potential impact of data centers on drought.

Project Structure and Relevant Datasets

Data Center Locations

Find location and basic information about data centers in the United States

- ➢ FracTracker Alliance U.S Data Center Tracker
- ➢ Brockovich AI Data Center Reporting (awaiting response)
- ➢ AI Data Center Community Tracker (awaiting response)

Environmental Data

Determine areas where data centers may put additional strain on the power grid and water supply

- ➢ EIA Electricity Generation
- ➢ U.S. Drought Monitor Data from 2006 - 2026

Health Data

Draw comparisons between various public health datasets from before and after the “AI Boom”

- ➢ America’s Health Rankings Annual Report
- ➢ CDC’s National Health Interview Survey
- ➢ SAMHSA’s National Survey on Drug Use and Health

Political Data

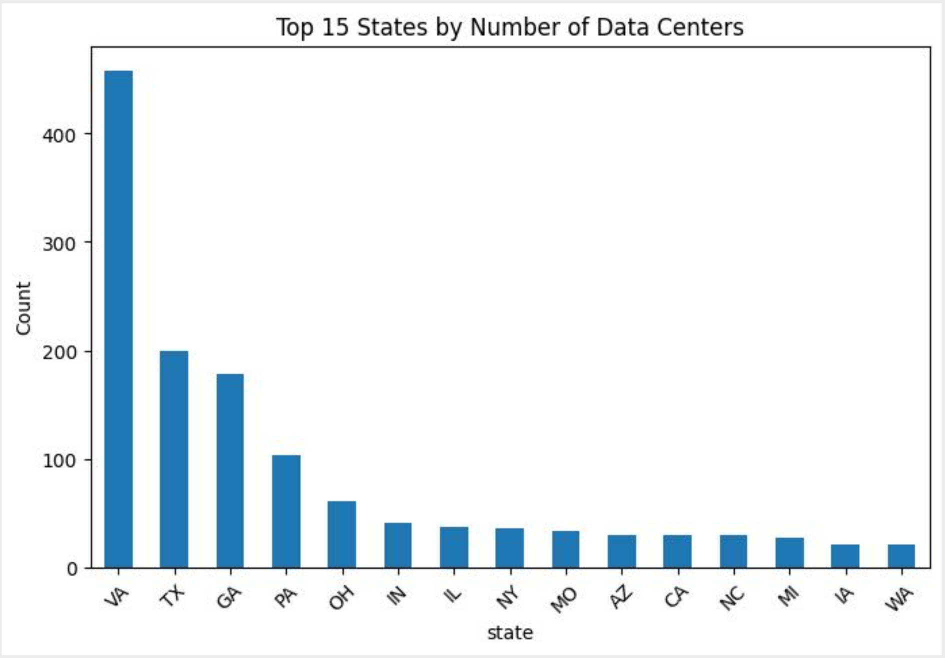
Explore location of data centers in red or blue districts and potential for community opposition

- ➢ MIT Election Data + Science Lab datasets on precinct-level returns for 2024 state and federal elections
- ➢ American Local Government Elections Database
- ➢ National Neighborhood Data Archive

Data Inspection: Data Centers Data

FracTracker Alliance U.S. Data Centers Tracker

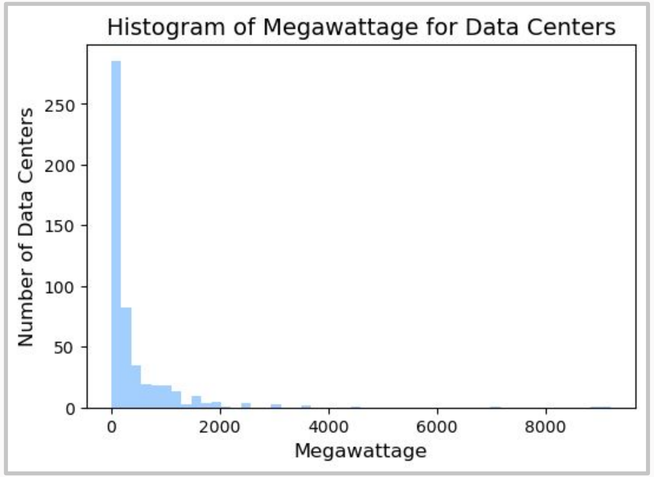
- 1592 rows, 44 columns
- Open Access
- Includes:
  - Data center locations (including lat/long), covering 873 U.S. zip codes
  - Operator (e.g., Microsoft, Amazon)
  - Status (proposed, under construction, operating, etc.)
  - Size (physical and power demand)
- Will combine with data from other sources on data centers if permission received

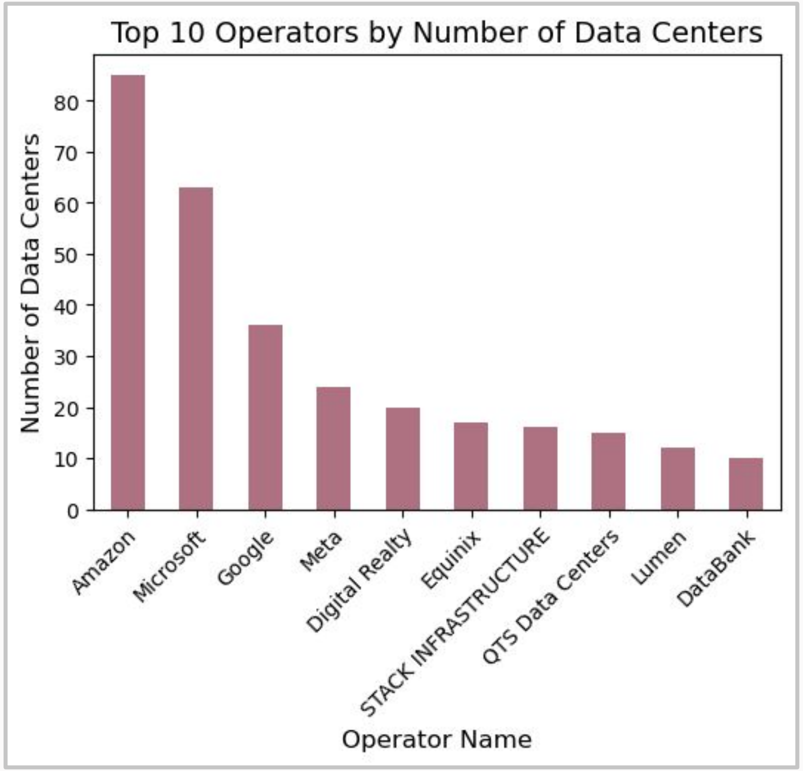


Data Inspection: Data Centers Data

Planned EDA

- Histograms/bar graphs to understand distributions of numerical and categorical features
- Investigate correlations between features
- Use in combination with following topic-specific datasets

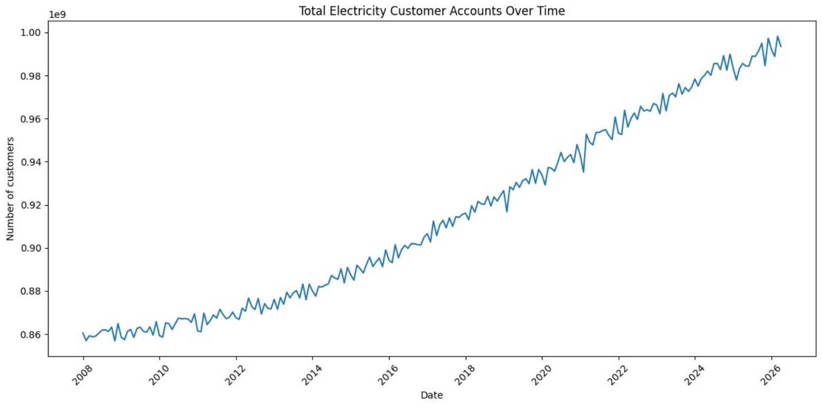


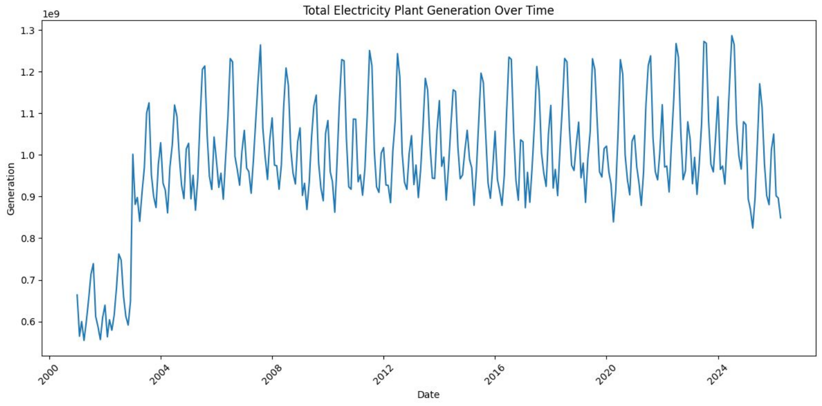


Data Inspection: Environmental Data

US Energy Information Administration Electricity Dataset

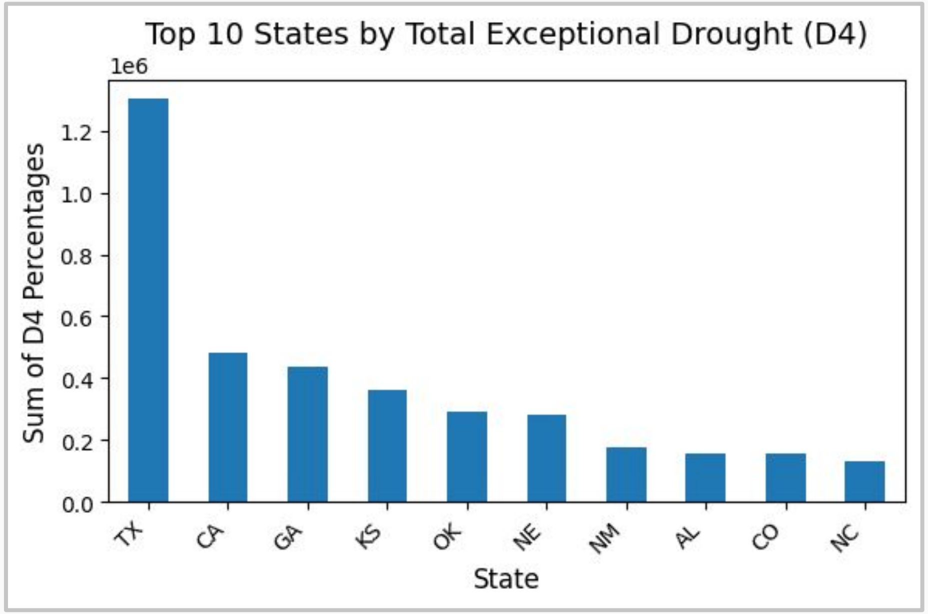
- Over 700,000 json files (all concatenated together into a single txt file)
- Open Access
- Includes:
  - Electricity consumption and generation
  - Region and state
  - Customer count
  - Sector (residential, commercial, etc)
  - Organized by year, quarter and month





Data Inspection: Drought Data

- Data Overview
  - Data separated by County, State
  - Contains:
    - D0: Abnormally Dry
    - D1: Moderate Drought
    - D2: Severe Drought
    - D3: Extreme Drought
    - D4: Exceptional Drought

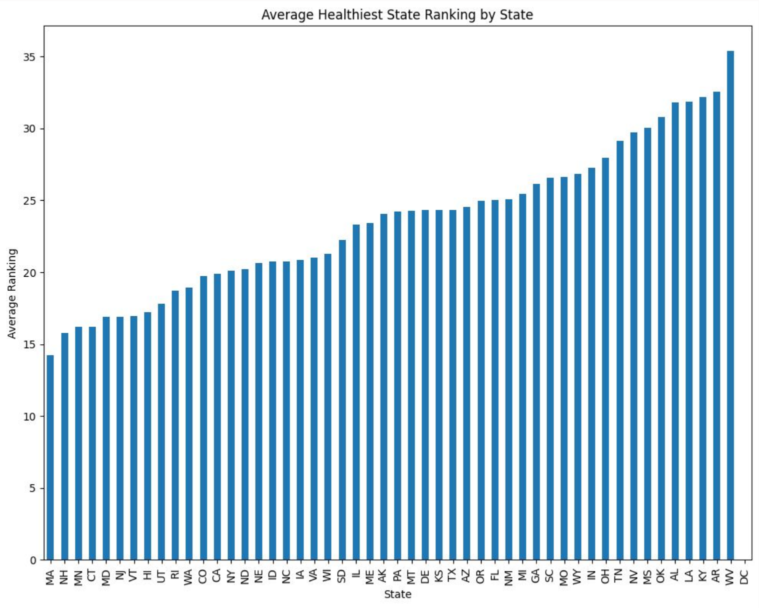


- Future Work
  - Want to look at drought over time
  - Want to graph on a US map

Data Inspection: Health Data

America's Health Rankings 2025 Report
- 80476 rows, 10 columns
- Includes state, rank, metric, and confidence interval
- Most included data are from 2023 onwards

CDC's National Health Interview Survey
- 32629 rows, 630 columns
- Will require extensive preprocessing and cleaning
    - Heavy reliance on codebook to determine which columns are pertinent to our objectives



Data Inspection: Health Data

SAMHSA's National Survey on Drug Use and Health

- Incredibly large dataset with availability only as SAS, SPSS, R, etc. files
- Also currently undergoing extensive preprocessing
  - Variables were edited, imputed, and recoded
  - Upwards of 75,000 study participants, 100+ columns

Future work

- Figure out how to best work with limited geographic information
  - Often only provides region (ex: "Northeast") instead of specific State
  - Need to confirm all datasets draw the same borders
- Continue to clean and preprocess all datasets
- EDA to investigate:
  - State-level illness/death rates
  - Pressing health concerns and increasing mental health problems in broader areas

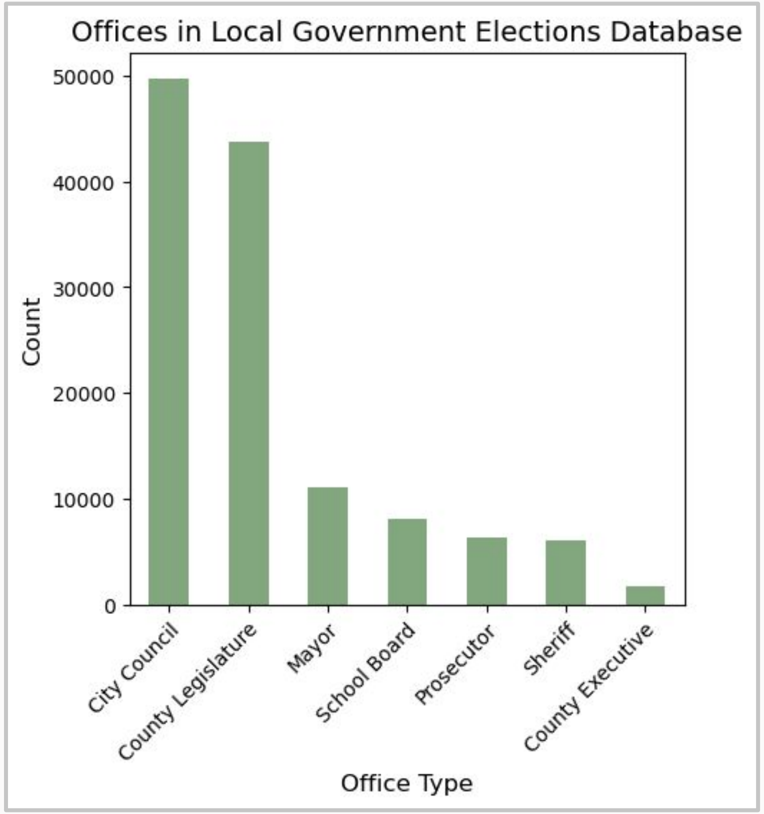
Data Inspection: Political Data

MIT Election Data + Science Lab 2024 precinct-level returns

- Voting results for state elections, U.S. Senate, House of Representatives, and President broken down by precinct and candidate
- Covers varying amount of precincts depending on subject

American Local Government Elections Database

- Local election results from 1989 to 2021
- Covers all states and 2,632 FIPS codes



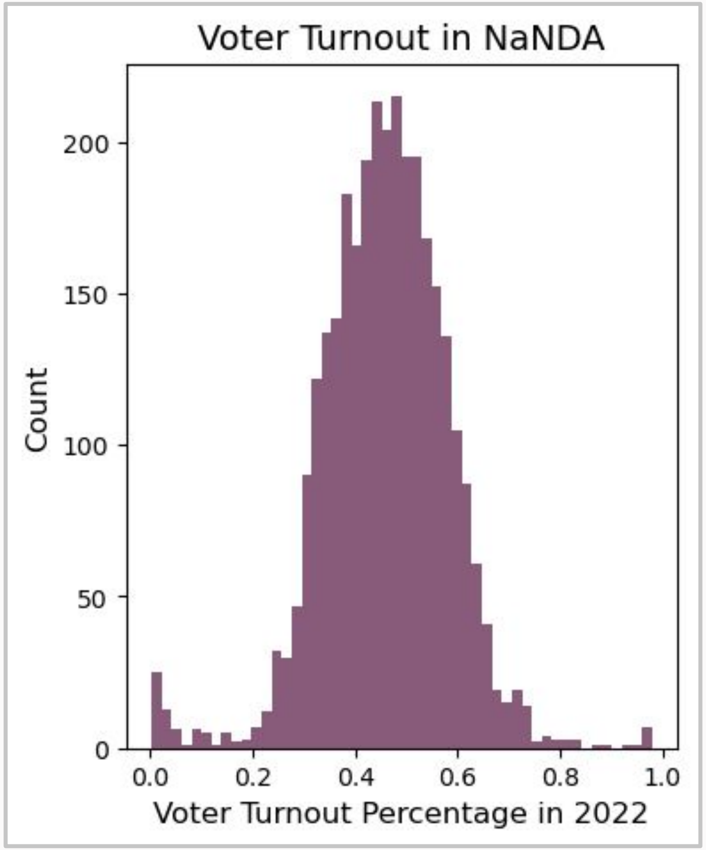
Data Inspection: Political Data

National Neighborhood Data Archive (NaNDA)

- FIPS code level information on voter turnout and long-term partisanship up to 2022
- Covers 3,108 FIPS codes

Future Work

- Develop strategies for tying together location to merge with data centers data
- Find correlations between partisanship and data center siting
- Find correlations between voter turnout and data center siting; could imply amount of potential opposition



Summary of Next Steps

- Finalize list of data sources to draw from
    - Potential to add more on e.g. basic demographic information
- Continue EDA to better understand individual datasets
- Continue to clean and preprocess necessary datasets
- Develop strategy for joining datasets
    - Mainly location-based
- Analyze correlations between data center siting and environmental, health, and political factors through EDA
    - Apply techniques like statistical analysis, investigation of varying visualization types
- Create interactive dashboard with visualizations of findings

Thank you